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GitHub Link: <https://github.com/Meet-6226/Flutter-Assignment>

Library Management System – Dart

1. Project Overview

The Library Management System is a console-based application developed using Dart. It simulates basic library operations such as managing books and other media, registering members, searching the catalog, borrowing items, and returning them.

The project is mainly designed to demonstrate important Object-Oriented Programming (OOP) concepts in Dart, including abstraction, inheritance, polymorphism, encapsulation, collections, and control flow.

The system supports three types of library resources: Books, DVDs, and Magazines. All of them share common properties but also have their own specific attributes and behaviour.

2. Key Components and Class Structure

LibraryItem – Abstract Base Class

LibraryItem is the abstract base class for all resources in the library.

Attributes:

- id – Unique identifier of the item.
- title – Name of the item.
- year – Publication or release year.
- isBorrowed – Indicates whether the item is currently borrowed.

It provides a status getter that returns “Borrowed” or “Available”, along with a `displayInfo()` method for displaying basic item details.

Being an abstract class allows it to act as a common blueprint for different types of library resources.

Book

Book extends LibraryItem and adds:

- author
- genre
- pageCount

It overrides `displayInfo()` to display book-specific information.

DVD

DVD extends `LibraryItem` and contains:

- director
- durationMins

Its `displayInfo()` method is customised to display movie-related details.

Magazine

Magazine extends `LibraryItem` and adds:

- issue
- publisher

It also overrides `displayInfo()` to display information relevant to magazines.

The three subclasses demonstrate inheritance and polymorphism, as they reuse common functionality from `LibraryItem` while providing their own implementation of `displayInfo()`.

3. LibraryMember Class

The `LibraryMember` class represents a registered user of the library.

Attributes:

- memberId
- name
- borrowedItems – A list containing the items currently borrowed.

The `printSummary()` method displays the member's details and the titles of their currently borrowed items.

This class helps maintain a clear relationship between a library member and the resources they have checked out.

4. Library Class

The `Library` class acts as the main manager of the system.

Attributes:

- name – Library name.
- catalog – A List containing all library items.
- members – A `Map<String, LibraryMember>` containing registered members.

Main Methods

- `addItem()` – Adds an item to the library catalog.
- `registerMember()` – Registers a new member using their unique ID.
- `showCatalog()` – Displays all items in the catalog.
- `search()` – Searches for items by title using a case-insensitive query.

- `checkoutItem()` – Checks whether a member and item are valid and whether the item is available before issuing it.
- `returnItem()` – Removes the item from the member's borrowed list and changes its status back to available.

The use of a List for catalog management and a Map for member management demonstrates Dart's collection handling capabilities.

5. Program Workflow

The `main()` function demonstrates the complete working of the system.

First, the application creates a library named "Metropolitan Public Library". Sample books, a DVD, and magazines are then created and added to the catalog.

Three members are registered:

- Sarah Jenkins
- David Miller
- Emma Watson

The application then demonstrates the catalog search functionality by searching for items matching "tech".

Borrowing Process

Sarah successfully checks out 1984 and Interstellar. When David tries to borrow Interstellar, the system rejects the request because the DVD is already borrowed by Sarah.

Emma successfully checks out Design Patterns.

The system then displays each member's current borrowed items using `printSummary()`.

Finally, Sarah returns Interstellar. Its status changes back to Available, allowing David to successfully check it out.

This demonstrates the complete borrowing lifecycle:

Available → Borrowed → Returned → Available → Borrowed

6. OOP Concepts Demonstrated

The project demonstrates the following core concepts:

- Abstraction: `LibraryItem` provides a common blueprint for all library resources.
- Inheritance: `Book`, `DVD`, and `Magazine` inherit from `LibraryItem`.
- Polymorphism: Each subclass provides its own version of `displayInfo()`.
- Encapsulation: Data and related operations are organised within their respective classes.
- Collections: List and Map are used for catalog and member management.
- Control Flow: Conditions and validations are used to handle borrowing, returning, searching, and unavailable items.

